

TECHNICAL ROADMAP TEMPLATE



Submission Deadline: May 17, 2026

Format: PDF (Strictly maximum 5 pages)

Naming Convention: [Track]_[University]_[TeamName]_Roadmap.pdf

TEAM INFORMATION

Team Name	DNTU - Althena
Institution	Dong Nai Technology University (DNTU)
Country	Vietnam
Track	Climate Change
Team Leader Name	Hoang Minh Tuan Anh, hoangminhtuananh1652005@gmail.com , +84377308690

SECTION 1: EXECUTIVE SUMMARY (PROBLEM-SOLUTION FIT)

Climate Resilience Copilot addresses a concrete ASEAN climate-resilience challenge: extreme heat waves driven by climate change increase air-conditioning and cooling demand, causing electrical overload, wiring overheating, short circuits, and ultimately urban fires. The project focuses on early detection and prevention of electrical fire risk in buildings, dormitories, campuses, and small urban zones where grid stress rises rapidly during hot weather.

Our solution combines Matter and Zigbee IoT telemetry, FIWARE Orion digital twins, and an MCP-based AI orchestration layer to detect hazardous compound conditions in near real time. Temperature, active power, and humidity readings are normalized into NGSI-v2 entities, continuously evaluated by a rule-first risk engine, and then enriched by AI reasoning for human-readable rationale and recommended actions.

When the system detects warning or critical thresholds, it publishes alerts, visualizes risk on a monitoring dashboard, and can trigger controlled response actions such as power cut or alert signaling with ACK/ERROR tracking and audit trail. In the current MVP, command execution is implemented as a simulated command path that is extensible to real hardware bridges. The architecture targets ≤ 5 -second detection in the demo flow, aims for zero false negatives in critical scenarios, and supports safe expansion from one room or building to multi-zone deployments across ASEAN cities exposed to climate-driven heat stress.

SECTION 2: TECHNICAL ARCHITECTURE

2.1 System Components:

Inputs: The system receives real-time telemetry from Matter emulators and optional Zigbee devices, including temperature ($^{\circ}\text{C}$), active power (W), humidity (%RH), device status (onOff), timestamps, and zone context. It can also receive operator requests through the dashboard and Telegram/OpenClaw gateway for status queries, manual evaluation, and controlled commands.

Processing Core: The AI happens inside the MCP Agent layer (Node.js, port 3002), where a rule-based risk engine computes the baseline risk first, then an AI reasoner enriches the rationale and recommended actions. Context is stored centrally in FIWARE Orion (NGSI-v2), while FIMAT Agent and

Zigbee Bridge normalize telemetry into digital twins. Proxy and dashboard services expose the live state for human monitoring.

Outputs: The system delivers risk scores, risk levels, natural-language rationale, dashboards, alert events, command executions with ACK/ERROR tracking, and optional multi-channel notifications via the OpenClaw Telegram gateway. In the current MVP baseline, ACK/ERROR results are generated by the command-executor simulation unless a real hardware bridge is connected.

2.2 Architecture Diagram:

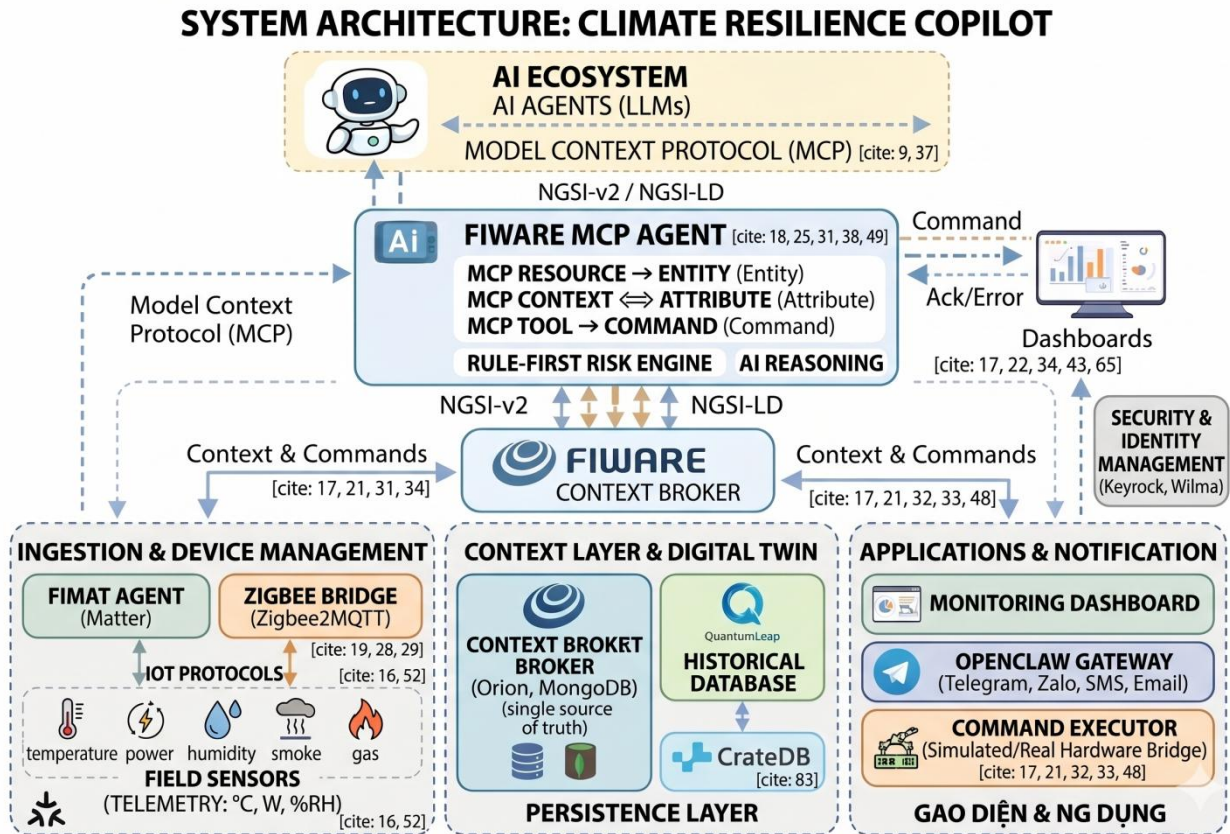


Figure 1. Reference architecture concept for integrating AI Agents (LLMs), MCP, FIWARE Context Broker, IoT Agents, applications, and dashboards. In this project, the same principle is applied through FIMAT Agent / Zigbee Bridge -> Orion -> MCP Agent -> Dashboard / Action / Notification.

Architecture details derived from the source code and docs:

- Devices and Emulators: matter-emulators simulate TemperatureSensor, HumiditySensor, and SmartPlug telemetry.
- Ingestion Layer: FIMAT Agent (port 3000) converts Matter events to NGSI-v2 and upserts entities into Orion. Zigbee Bridge (port 3003) performs the same role for zigbee2mqtt and MQTT devices.
- Context Layer: FIWARE Orion Context Broker (port 1026, backed by MongoDB 4.4) is the single source of truth for digital-twin state.
- Decision Layer: MCP Agent (port 3002) polls Orion every 5 seconds, computes risk, enriches explanations with AI, publishes AlertEvent and ZoneRisk entities, and executes protective commands when policy allows.
- Visualization Layer: monitor-dashboard (port 8080) displays devices, alerts, maps, simulator controls, and action visibility through a CORS proxy (port 3001).
- Notification Layer: OpenClaw Gateway (port 3004) provides Telegram-based routing, alert push, and operational skill shortcuts.

Core non-negotiable system flow: Device/Event -> FIMAT -> Orion -> Dashboard/MCP -> Action/Ack.

SECTION 3: AI APPROACH & MODEL SELECTION

Primary AI Approach: [] Machine Learning [] NLP [] Computer Vision [X] Generative AI [X]

Other: Rule-based risk engine and digital twin orchestration

Model Selection: The repository is designed so the MCP Agent can work with an optional AI endpoint through environment variables (AI_ENDPOINT, AI_API_KEY, AI_MODEL; current default in config is gpt-4o-mini if configured). However, the project is intentionally rule-first: critical detection does not depend on the AI model. The AI layer is used mainly for reasoning quality, explanation, and action recommendation. Core frameworks and runtime already implemented in source code include Node.js, Express, Axios, MQTT, FIWARE Orion, MongoDB, and optional Telegram bot integration.

Reasoning: This is the most efficient approach because life-safety detection needs determinism, low latency, and graceful degradation. The implemented risk engine already combines temperature, active power, and humidity thresholds. If AI is unavailable, the system still detects critical risk and publishes alerts through deterministic rules, while command handling continues through the MVP command path. When AI is available, it improves explanation quality and operator trust without becoming a single point of failure.

- Implemented warning thresholds: temperature $\geq 40^{\circ}\text{C}$, activePower $\geq 800\text{W}$, humidity $\geq 75\%$.
- Implemented critical thresholds: temperature $\geq 50^{\circ}\text{C}$, activePower $\geq 950\text{W}$, humidity $\geq 90\%$ combined with high power.
- Critical recommended actions defined in the code logic include CUT_POWER_IMMEDIATELY, TURN_ON_ALERT_LAMP, NOTIFY_OPERATOR, and EVACUATE_ZONE. In the current MVP, this action list represents the intended response policy, while hardware execution remains partially simulated.
- Implemented MCP endpoints include /risk, /risk/all, /tools/query_entities, /tools/compute_risk, /tools/publish_alert, /tools/invoke_command, /alerts, /commands, /evaluate, and /tools/ai_chat.

SECTION 4: DATA STRATEGY & ETHICS

(Based on Workshop 2: Technical Architecture and Data Ethics)

Data Sources: The MVP uses three primary data sources: (1) synthetic telemetry from Matter emulators in the repository, (2) optional live Zigbee devices via zigbee2mqtt/MQTT, and (3) optional WiFi/Tuya integration described in docs/IOT_DEVICE_GUIDE.md. All data is operational telemetry only: temperature, humidity, power, switch status, timestamps, zone, and device identifiers.

Data Quality and Cleaning: Data quality is enforced through normalized NGSI-v2 entities, stable entity IDs, per-attribute timestamp metadata, unit metadata, serialized per-entity event processing in FIMAT Agent, idempotent upserts, and append fallback when attributes do not yet exist. The build and test gate also requires smoke tests, a demo scenario, dashboard availability, and validation that SmartPlug contains both onOff and activePower attributes.

Licensing and Legality: The software stack relies on open technologies and packages already declared in the repository, including Node.js packages, FIWARE Orion, MongoDB images, MQTT tooling, and Telegram bot libraries. The roadmap avoids any restricted private datasets by default because the demo can run entirely on synthetic and self-generated telemetry. This keeps the hackathon setup simple and legally safe.

Bias Mitigation and Fairness: This solution minimizes classic demographic bias because it does not process personal identity data, faces, voice, or sensitive personal records. The MVP explicitly avoids PII and uses only device and zone telemetry. Fairness risks instead come from infrastructure inequality, sensor placement, and uneven environmental exposure across buildings or regions. To mitigate that, the system uses transparent rules, explicit thresholds, explainable rationale, human-in-the-loop for sensitive actions, and audit trails for every command decision.

- No PII in MVP; only telemetry device and zone context is used.
- Rule-first safety design reduces opaque AI-only decision making.
- Audit trail is mandatory for actions: who, when, why, and result.

Sensitive actions require human oversight when deployed beyond controlled demo scope.

SECTION 5: DEVELOPMENT MILESTONES (AGILE ROADMAP)

(Based on Workshop 3: Agile Prototyping)

The roadmap below is aligned to the actual repository status and the hackathon objective of preparing a stable semi-final prototype.

Phase	Activity / Task	Tools Used	Expected Outcome
Sprint 1 (Completed /	Establish FIWARE	Docker Compose,	Working end-to-end

Current Baseline)	foundation: Docker Compose for Orion and MongoDB, FIMAT Matter-to-NGSI ingestion, proxy server, dashboard, smoke scripts, entity model, and API docs.	Node.js, Express, FIWARE Orion, MongoDB, PowerShell	MVP baseline with live entities in Orion and dashboard visibility.
Sprint 2	Harden risk detection: refine MCP risk-engine thresholds, validate critical heat-wave and moisture-short-circuit scenarios, enforce alert dedupe and cooldown, improve rationale consistency.	MCP Agent, rule engine, demo-scenario.ps1, smoke-test.ps1, Orion entities	Reliable warning and critical detection within <= 5 seconds and zero false negative in critical demo paths.
Sprint 3	Strengthen action and operator workflow: command ACK/ERROR tracking, alert lamp and smart plug control, OpenClaw Telegram integration, clearer action visibility in dashboard.	Command executor, OpenClaw Gateway, Telegram Bot API, dashboard modules	Demonstrable closed-loop response: detect -> alert -> act -> acknowledge -> audit.
Sprint 4	Production-readiness for hackathon judging: refine docs, FAQ, KPI scorecard, test report, architecture narrative, and replayable 3-5 minute demo.	README_HACKATHON, ARCHITECTURE.md, KPI_SCORECARD.md, DEMO_SCRIPT.md, TEST_REPORT.md	Judge-ready prototype with reproducible setup, stable storyline, and measurable impact metrics.

SECTION 6: SCALABILITY & REGIONAL RESILIENCE

Scaling Potential: The architecture already separates ingestion, context, decision, visualization, and notification layers. This makes it easy to add more buildings, campuses, or districts as new zones without rewriting the whole system. FIWARE Orion provides a unified digital-twin interface, while the MCP Agent can reason over multi-zone context consistently. The same pattern can extend from one campus in Vietnam to public housing, industrial parks, universities, and municipal pilot deployments across ASEAN countries facing heat-wave stress and electrical fire risk.

Technical Constraints: The current prototype is still an MVP. It uses emulators for the default demo, lacks full production security and HA, depends on polling rather than full event subscriptions, and has limited long-term historical storage and analytics. AI integration is optional and environment-driven, which is good for resilience but means explanation quality depends on external model availability if enabled. Hardware reliability, MQTT infrastructure, and regional device standards must be validated separately in field deployment.

- Regional deployment path: 1 room -> 1 building -> 1 campus -> 1 district -> multi-city network.
- Multi-protocol support already planned: Matter, Zigbee, optional Tuya and WiFi integration.
- Clear extension path for multi-channel alerts: Telegram, Zalo, SMS, and Email.
- Future resilience features: historical analytics, replay timeline, CI smoke tests, richer policy control, and additional sensors such as smoke, gas, motion, and water leak.

INSTRUCTIONS FOR SUBMISSION:

1. **Strictly PDF:** Convert your final document to PDF.
2. **Visuals Matter:** Use clear diagrams for Section 2.

3. **Submission Link:** Team Leaders must upload this file to the **Official Submission Folder** [SUBMISSION - TECHNICAL ROADMAP](#) by **May 17, 2026** (9:00 PM GMT+7).